

Name: Calvin, Andrew, Dylan **Date:** 5/8/2024 **Class:** Engineering Period 3

Design Process Reflection

- The class met with the main stakeholder that runs the TV department at our highschool, and he ran through a list of things that needed fixing and that we could design and create stuff for. We chose to design and create our own lens covers, as the TV department has lost a few over the years. The problem statement was that the camera lens on the cameras can collect dust and fingerprints on the glass, and if they are dropped, they can get scratches on the lens. This worsens the video quality, and it is very expensive to replace the lens. So, when you're not using it, it is wise to have a camera lens cover to place over it so that it cannot get damaged or dirty.
- The first thing we did was define what the problem actually was. We described the problem in detail, looked into designs that were already created, and finally defined our criteria that were necessary to solve the problem. The main thing that was a problem for us was finding the necessary field experts to review our products. The only field expert we were able to find at first was Mr. O'Farrell, after which we struggled to find people in the necessary fields to see if what we created was good enough. Information about lens covers online was also difficult to look for, since this is a relatively niche product. Not many people have made unique designs for lens covers in the past, which made it difficult to find any reference points when making our own cover. At least defining our criteria was somewhat easier, as we knew what we needed our design to do in order to be successful. Our cover's main goals were to protect the camera lens, while remaining durable and not falling off. These were all talked about in the design brief. Once we were able to pin down our main objectives, we were able to describe what they were and why they were so important.
- Next came the ideation phase, where we went over the ideas we came up with to solve the problem. We came up with 10 ideas at first, but for this step, we had to narrow our pool of ideas down to 3. For each of these ideas, we went into detail about the dimensions of our designs, their specific features, and how they would be constructed (i.e. the cost of materials). The difficult part was deciding what the priority of our criteria would be, as it wasn't immediately obvious what aspect of our design was more important than the others. However, the rest of this phase was rather easy, as we were familiar enough with our own ideas to know how they would be printed and describe how they would work.

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- After this, we entered the selection and prototype design phase. Here, we had to choose which design from element D we would move forward with. Our choices were a cover with a door that slides back and forth, a cover that clicks into place on the lens, and a cover made of felt that fits around the cover flexibly. It was difficult to choose which of these to pick from, and as a result of that, we never really stuck with a single design until the very end. We kept switching between the sliding door and slide-on covers, even coming up with the idea to combine them at a later point, until we eventually stuck with the idea of the slide-on cover. If we were to do this project again in the future, we would agree on a design sooner to prevent this from happening again.
- After this came the actual construction of the prototype. We went over all of our previous iterations of our designs, including the changes between versions. We created a CAD design of our first concept ideas, and then we printed it out. This design had 4 prongs to hold the lens in place, however two of these were getting in the way and didn't let the lens cover attach. So, we cut them off, and then attached the lens cover with the other two prongs, and it fit well and held strong. Then, we made some small changes to make it more ergonomic. Then, we increased the durability to make sure that it wouldn't break easily out in the field. After this, we made the final design and printed it out.
- We also carried out multiple tests on the product. The first was a durability test to make sure that the camera lens cover wouldn't break easily. We dropped it from 9 feet (the ceiling) 5 times in a row, and it ended up getting cracks and the 3d printing of the outer layer was peeling off. We adjusted the 3d print settings to make the outer walls thicker to stop this from happening. We tested it again and it passed. We also had a shake test to make sure it held on to the camera securely. We violently shook it, but it held firm. We also did smaller tests that just made sure we were checking our requirements, like having the cover attached to the camera when not on the lens, and the weight being under 0.5 grams. It also had to go on and off easily, which it does, as it only takes about a second to slide it on and off.
- The project for this group went well because we were able to come up with good ideas early on in order to create a good groundwork for the product. The designing of the CAD was interesting and challenged us, but in the end it turned out great. The lens cover functions well and is easy to use. I think it could be a legitimate product if it was actually produced. Overall, the creation of the product went well.